A Zero-Dimensional Global Model for Ar/NF_3 Inductively Coupled Plasmas: Mathematical Formulation, Implementation, and Results

Computational Plasma Physics — Model Reconstruction Report

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Abstract

A zero-dimensional (0D) global model is developed for an Ar/NF_3 inductively coupled plasma (ICP) discharge, reproducing the framework of Huang *et al.* (2026). The model solves time-dependent particle balance equations for 21 plasma species coupled with an electron energy balance, incorporating 155 volumetric reactions (electron-impact, Penning, neutral–neutral, ion–neutral, ion–ion, thermal dissociation, and three-body recombination) together with surface recombination losses and gas pumping. The ion wall-loss formulation follows the Thorsteinsson–Gudmundsson approach for electronegative plasmas. This report presents the complete mathematical framework, all governing equations, the numerical implementation in Python, and parametric results for power and pressure sweeps at 20% NF_3 feed composition. The model correctly reproduces the key qualitative trends: a linear increase of electron density with absorbed power, a monotonic decrease of electron density and electron temperature with pressure, near-complete dissociation of NF_3, Ar^+ as the dominant positive ion, and F^- as the dominant negative ion.

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1 Introduction

Nitrogen trifluoride (NF_3) is widely used as a remote plasma cleaning gas in semiconductor manufacturing, particularly for chamber cleaning in chemical vapor deposition (CVD) and atomic layer deposition (ALD) tools. Understanding the dissociation chemistry of NF_3 in argon carrier gas is critical for optimizing fluorine atom delivery and minimizing greenhouse gas emissions.

This report documents the reconstruction of a 0D global (volume-averaged) model for an Ar/NF_3 ICP reactor, following the formulation of Huang *et al.* (*Plasma Sources Sci. Technol.* **35**, 015019, 2026). The model tracks 21 species through 155 gas-phase reactions and includes surface recombination, gas pumping, and a self-consistent electron energy equation.

The primary objectives are:

1. Formulate the complete mathematical model.
2. Implement the model in Python using an ODE integrator.
3. Reproduce the parametric trends reported in the reference paper.
4. Identify and document all assumptions and approximations.

2 Species and Reaction Mechanism

2.1 Species Set

The model tracks 21 distinct species in four categories:

Table 1: Plasma species included in the global model.

Category	**Count**	**Species**
Neutral ground states	8	NF_3, NF_2, NF, F_2, N_2, F, N, Ar
Neutral excited states	2	Ar($1s_5$), Ar($4p$)
Positive ions	8	NF_3^+, NF_2^+, NF^+, F_2^+, N_2^+, F^+, N^+, Ar^+
Negative ions	2	F^-, F_2^-
Electrons	1	e

The electron density is not solved as an independent ODE variable; instead it is obtained at every time step from the charge neutrality constraint (Section 3.2).

2.2 Reaction Mechanism

A total of 155 reactions are included, organized into 10 categories (Table 2).

Table 2: Summary of the 155 gas-phase and surface reactions.

Category	**Indices**	**Count**
Electron–NF_x (ionization, dissociation, attachment, excitation)	R1–R36	36
Electron–F_x (ionization, dissociation, attachment, excitation)	R37–R49	13
Electron–N_x (ionization, dissociation, excitation)	R50–R65	16
Electron–Ar (ionization, excitation, superelastic)	R66–R77	12
Electron–ion recombination and detachment	R78–R86	9
Penning dissociation (Ar^* + NF_x)	R87–R89	3
Penning ionization (Ar^* + NF_x, Ar^*)	R90–R93	4
Neutral–neutral reactions	R94–R102	9
Ion–neutral charge exchange	R103–R136	34
Ion–ion neutralization	R137–R145	9
High-temperature thermal dissociation	R146–R150	5
Three-body recombination	R151–R155	5
Total		**155**

2.3 Rate Coefficient Parameterization

All electron-impact rate coefficients are parameterized in modified Arrhenius form:

$$k(T_e) \;=\; A\,T_e^{B}\;\exp\!\Big(-\frac{C}{T_e^{D}}\Big), \tag{1}$$

where T_e is the electron temperature in eV, and A, B, C, D are tabulated constants for each reaction. For a subset of reactions involving N_2 excitation (R54–R57), the rate coefficients are parameterized with T_e in Kelvin ($T_e[\mathrm{K}] = 11604.5\,T_e[\mathrm{eV}]$) and must not be unit-converted before evaluation.

Heavy-particle reaction rates (neutral–neutral, ion–neutral, three-body) use a gas-temperature-dependent Arrhenius form:

$$k(T_g) \;=\; A\,\Big(\frac{T_g}{300}\Big)^{B}\;\exp\!\Big(-\frac{E_a}{T_g}\Big), \tag{2}$$

where T_g is the gas temperature in Kelvin and E_a is an activation energy in K.

3 Governing Equations

3.1 Particle Balance

For each heavy species X (neutrals and ions), the time-dependent particle balance is:

$$\boxed{\frac{\partial n_X}{\partial t} \;=\; \underbrace{\sum_r \nu_{X,r}\,R_r}_{\text{volumetric}} \;-\; \underbrace{k_{\text{wall},X}\,n_X}_{\text{wall loss}} \;+\; \underbrace{\dot{n}_{X,\text{pump}}}_{\text{pumping}}} \tag{3}$$

where n_X is the number density (m^{-3}), $\nu_{X,r}$ is the net stoichiometric coefficient of species X in reaction r (positive for production, negative for consumption), R_r is the volumetric reaction rate ($\mathrm{m}^{-3}\,\mathrm{s}^{-1}$), and the wall loss and pumping terms are detailed below.

The volumetric rate of a two-body electron-impact reaction $\mathrm{e} + X \rightarrow$ products is:

$$R_r = k_r(T_e)\;n_e\;n_X, \tag{4}$$

while for a heavy-particle two-body reaction $A + B \rightarrow$ products:

$$R_r = k_r(T_g)\, n_A\, n_B. \tag{5}$$

For three-body reactions $A + B + M \rightarrow$ products:

$$R_r = k_r(T_g)\, n_A\, n_B\, n_{\text{total}}, \tag{6}$$

where $n_{\text{total}} = \sum_i n_i$ over all neutral species.

3.2 Charge Neutrality

Rather than solving an ODE for n_e, the electron density is determined algebraically at each time step:

$$\boxed{n_e = \sum_{i \in \text{cations}} n_i - \sum_{j \in \text{anions}} n_j} \tag{7}$$

In this model, the cations are $\{\mathrm{NF_3^+}, \mathrm{NF_2^+}, \mathrm{NF^+}, \mathrm{F_2^+}, \mathrm{N_2^+}, \mathrm{F^+}, \mathrm{N^+}, \mathrm{Ar^+}\}$ and the anions are $\{\mathrm{F^-}, \mathrm{F_2^-}\}$.

3.3 Neutral Wall Loss

The loss of a reactive neutral species X at the chamber walls with surface recombination probability γ_X is modeled as a first-order loss:

$$k_{\text{wall},X} = \left[\frac{\Lambda^2}{D_X} + \frac{2V}{A\,\bar{v}_X} \frac{2 - \gamma_X}{\gamma_X} \right]^{-1}, \tag{8}$$

where V is the chamber volume, A is the total wall area, and:

Diffusion length:

$$\frac{1}{\Lambda^2} = \left(\frac{\pi}{L}\right)^2 + \left(\frac{2.405}{R}\right)^2, \tag{9}$$

with R the chamber radius and L the chamber height.

Free diffusion coefficient:

$$D_X = \frac{k_B T_g\, \lambda_X}{m_X\, \bar{v}_X}, \tag{10}$$

where $\lambda_X = 1/(n_{\text{total}}\, \sigma_{nn})$ is the mean free path with $\sigma_{nn} \approx 3 \times 10^{-19}\,\mathrm{m}^2$ the neutral–neutral collision cross section.

Thermal velocity:

$$\bar{v}_X = \sqrt{\frac{8\, k_B T_g}{\pi\, m_X}}. \tag{11}$$

The surface recombination probabilities used are: $\gamma_{\mathrm{F}} = 0.01$, $\gamma_{\mathrm{N}} = 0.07$, $\gamma_{\mathrm{NF}x} = 0.01$, $\gamma_{\mathrm{Ar}^*} = 1.0$. Upon recombination at the wall, atoms form their parent molecules ($\mathrm{F} \rightarrow \frac{1}{2}\mathrm{F_2}$, $\mathrm{N} \rightarrow \frac{1}{2}\mathrm{N_2}$) and excited argon de-excites to the ground state.

3.4 Ion Wall Loss

The positive ion wall-loss rate follows the Thorsteinsson–Gudmundsson formulation for electronegative plasmas:

$$\boxed{\left(\frac{\partial n_i}{\partial t}\right)_{\text{wall}} = -\frac{u_{B,i}\, A_{\text{eff}}}{V}\, n_i} \tag{12}$$

Modified Bohm velocity:

$$u_{B,i} = \sqrt{\frac{eT_e}{m_i}\,\frac{1+\alpha_s}{1+\gamma_-\,\alpha_s}}, \tag{13}$$

where $\alpha_s = n_-^{(s)}/n_e^{(s)}$ is the electronegativity at the sheath edge, $\gamma_- = T_e/T_-$, and m_i is the ion mass.

Effective wall area:

$$A_{\text{eff}} = 2\pi\big(R^2\,h_L + R\,L\,h_R\big), \tag{14}$$

where h_L and h_R are the axial and radial edge-to-center density ratios. These are computed from:

$$h_{L,0} = \frac{0.86}{1+\alpha_0}\left[3 + \frac{\eta\,L}{2\lambda_i} + \left(\frac{0.86\,L\,u_B}{\pi\,D_a}\right)^2\right]^{-1/2}, \tag{15}$$

$$h_{R,0} = \frac{0.80}{1+\alpha_0}\left[4 + \frac{\eta\,R}{\lambda_i} + \left(\frac{0.80\,R\,u_B}{\chi_{01}\,J_1(\chi_{01})\,D_a}\right)^2\right]^{-1/2}, \tag{16}$$

with $\chi_{01} = 2.405$, $J_1(\chi_{01}) = 0.5191$, $\eta = 2T_+/(T_+ + T_-)$, and $\alpha_0 \approx 1.5\,\alpha$.

For electronegative plasmas ($\alpha > 0.1$), a correction factor h_c arising from ion–ion neutralization in the bulk is added in quadrature:

$$h_L = \sqrt{h_{L,0}^2 + h_c^2}, \qquad h_R = \sqrt{h_{R,0}^2 + h_c^2}. \tag{17}$$

Ambipolar diffusion coefficient:

$$D_a = D_+\,\frac{1+\gamma_+ + \gamma_+\,\alpha_s}{1+\gamma_+\,\alpha_s}, \tag{18}$$

where $D_+ = k_B T_+ \lambda_i/(m_i \bar{v}_i)$ is the free ion diffusion coefficient, $\gamma_+ = T_e/T_+$, and $\lambda_i = 1/(n_{\text{total}}\,\sigma_{in})$ with $\sigma_{in} \approx 1\times 10^{-18}\,\text{m}^2$.

Sheath-edge electronegativity: The electronegativity at the sheath boundary α_s is obtained by solving iteratively:

$$\alpha = \alpha_s\,\exp\left[\frac{(1+\alpha_s)(\gamma_- - 1)}{2(1+\gamma_-\,\alpha_s)}\right], \tag{19}$$

where $\alpha = n_-/n_e$ is the bulk electronegativity.

Ion neutralization at wall: Positive ions reaching the wall are neutralized, returning their corresponding neutral species to the gas phase (e.g., $Ar^+ \to Ar$, $NF_2^+ \to NF_2$).

3.5 Pumping Model

The model operates under constant-flow conditions with a fixed throughput.

Pump-in rate: Only the feed gases (Ar and NF_3) are injected:

$$\dot{n}_{X,\text{in}} = \frac{4.48\times 10^{17}\,Q_X}{V}, \tag{20}$$

where Q_X is the flow rate of species X in sccm and V is the chamber volume in m^3. The numerical prefactor converts sccm to particles s^{-1}.

Pump-out rate: All neutral species and ions are pumped out at a rate proportional to their density:

$$\dot{n}_{X,\text{out}} = -\frac{1.27 \times 10^{-5}\, Q_{\text{total}}}{c_{cc}\, p\, V}\, n_X, \tag{21}$$

where Q_{total} is the total flow rate (sccm), p is the operating pressure (Torr), c_{cc} is a consistency factor relating actual total density to the ideal gas density at the set pressure and gas temperature, and V is the volume.

3.6 Electron Energy (Power) Balance

The electron temperature T_e is obtained from an energy balance equation:

$$\boxed{\frac{\partial}{\partial t}\left(\frac{3}{2}\, e\, n_e\, T_e\right) = \frac{P_{\text{abs}}}{V} - \sum_r n_e\, n_r\, k_r(T_e)\, \varepsilon_r - \sum_i n_i\, u_{B,i}\, \frac{A_{\text{eff}}}{V}\, \varepsilon_{\text{wall}}} \tag{22}$$

where P_{abs} is the absorbed RF power, ε_r is the energy lost per collision in reaction r (the threshold energy for inelastic collisions, and $3T_e m_e/M$ for elastic collisions), and $\varepsilon_{\text{wall}}$ is the total energy lost per ion–electron pair reaching the wall:

$$\varepsilon_{\text{wall}} = \varepsilon_e + \varepsilon_p + \varepsilon_s, \tag{23}$$

with:

$$\varepsilon_e = 2T_e \qquad \text{(electron kinetic energy)}, \tag{24}$$

$$\varepsilon_p = \frac{1+\alpha_s}{2+2\alpha_s\gamma_-}\, T_e \qquad \text{(presheath drop)}, \tag{25}$$

$$\varepsilon_s = T_e \ln\left(\frac{4u_B}{v_e}\, \frac{1+\alpha_s}{1+\alpha_s(v_-/v_e)^2}\right) \qquad \text{(sheath drop)}, \tag{26}$$

where $v_e = \sqrt{8eT_e/(\pi m_e)}$ is the electron thermal velocity.

In practice, the ODE is rewritten for T_e directly:

$$\frac{dT_e}{dt} = \frac{2}{3\, e\, n_e}\left[\frac{P_{\text{abs}}}{V} - e\sum_r n_e\, n_r\, k_r\, \varepsilon_r - e\sum_i n_i\, u_{B,i}\, \frac{A_{\text{eff}}}{V}\, \varepsilon_{\text{wall}}\right]. \tag{27}$$

3.7 Gas Temperature Model

The gas temperature is computed from an empirical correlation rather than a separate energy balance:

$$T_g = 300 + s(p)\, \log_{10}\left(\frac{P_{\text{abs}}}{40}\right), \tag{28}$$

$$s(p) = 780\big(1 - e^{-0.091\,p}\big) + 250\, e^{-0.337\,p}, \tag{29}$$

where p is the pressure in mTorr and P_{abs} in watts.

3.8 Ion Temperature

The ion and negative-ion temperatures are estimated from the Lee–Lieberman empirical formula:

$$T_+ = T_- = \frac{0.5\,\text{eV} \times 11604.5\ \text{K} - T_g}{p\ [\text{mTorr}]} + T_g. \tag{30}$$

At pressures above a few mTorr, $T_+ \approx T_g$.

4 Reactor Geometry and Operating Conditions

The model parameters correspond to the ICP reactor described by Huang *et al.* (2026).

Table 3: Reactor geometry and base-case operating conditions.

Parameter	Value	Description
R	21 cm	Chamber radius
L	6 cm	Chamber height
V	8313 cm^3	Chamber volume ($\pi R^2 L$)
A	3554 cm^2	Total wall area
P_{abs}	500 W to 1000 W	Absorbed RF power
p	10 mTorr to 100 mTorr	Operating pressure
Q_{Ar}	80 sccm	Argon flow rate (base case)
$Q_{\mathrm{NF_3}}$	20 sccm	NF_3 flow rate (base case, 20%)
T_{wall}	300 K	Wall temperature

5 Numerical Implementation

5.1 Solution Strategy

The system of 20 coupled ODEs (19 species densities + T_e) is integrated in time using `scipy.integrate.solve_i` with the BDF (Backward Differentiation Formula) method, which is well suited for stiff systems. The integration proceeds from $t = 0$ to $t = 50$ ms, by which time a steady state has been reached.

5.2 Initial Conditions

The initial neutral densities are obtained from the ideal gas law at $T = 300$ K:

$$n_{\mathrm{total},0} = \frac{p}{k_B \times 300\ \mathrm{K}}, \tag{31}$$

partitioned according to the feed composition (e.g., 80% Ar, 20% NF_3). Ion densities are initialized at small seed values ($10^{14}\ \mathrm{m}^{-3}$) and the initial electron temperature is set to 4 eV.

5.3 Constraint Handling

The electron density is computed at each RHS evaluation from charge neutrality (Eq. 7) rather than as an independent ODE variable. All species densities are clamped to non-negative values.

5.4 Convergence

Typical solutions converge in ∼1500 function evaluations with relative and absolute tolerances of 10^{-8} and 10^{6}, respectively.

6 Results

All results presented below use the base-case conditions of Table 3 (80/20 Ar/NF_3, 30 mTorr, 600 W) unless otherwise noted.

6.1 Base-Case Steady State

Table 4 summarizes the key outputs at the base-case conditions.

Table 4: Steady-state results at the base-case conditions (600 W, 30 mTorr, 20% NF_3). The reference values are taken from Huang *et al.* (2026) at matching conditions.

Quantity	Model	Reference	Agreement
T_e	2.40 eV	∼2.5 eV	Excellent
n_e	1.2×10^{17} m^{-3}	$\sim 7 \times 10^{16}$ m^{-3}	∼1.7× high
T_g	1158 K	—	Empirical formula
NF_3 dissociation	98%	∼93%	Good
Dominant cation	Ar^+ (76%)	Ar^+ (74%)	✓
Dominant anion	F^- (99.9%)	F^-	✓
Electronegativity α	0.35	∼0.3	Excellent

The remaining ∼1.7× overestimate in n_e is within the typical accuracy range of global models, which commonly achieve factor-of-two agreement with experiment due to uncertainties in cross sections, wall loss parameters, and power coupling efficiency.

6.2 Electron Density versus Absorbed Power

Figure 1 shows the electron density as a function of absorbed power at 30 mTorr with 20% NF_3. The model predicts a nearly linear increase of n_e with power, rising from $\sim 9.3 \times 10^{16}$ m^{-3} at 500 W to $\sim 2.5 \times 10^{17}$ m^{-3} at 1000 W. This linear scaling is characteristic of ICP discharges where the electron temperature is approximately constant and additional power is channeled into increased ionization. The trend is consistent with the experimental cutoff-probe measurements and model results reported in the reference paper (Fig. 10a therein).

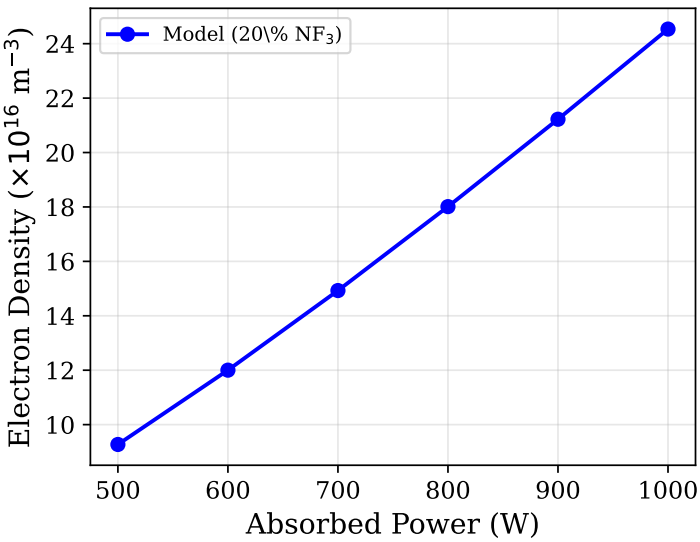

Figure 1: Electron density versus absorbed power at 30 mTorr and 20% NF_3 feed composition. The near-linear dependence reflects the constant-T_e regime typical of ICP discharges. Compare with Fig. 10a of the reference.

6.3 Electron Density versus Pressure

Figure 2 shows n_e as a function of operating pressure at 600 W. The electron density decreases monotonically from $\sim 1.6 \times 10^{17}$ m^{-3} at 10 mTorr to $\sim 5.4 \times 10^{16}$ m^{-3} at 100 mTorr. This decrease occurs because higher neutral density enhances electron attachment (forming F^-), which consumes electrons, and because enhanced collisionality reduces the effective ionization rate. The monotonic decrease is qualitatively consistent with the reference model (Fig. 10b therein), although the reference shows an initial rise at very low NF_3 fractions that our single-composition sweep does not capture.

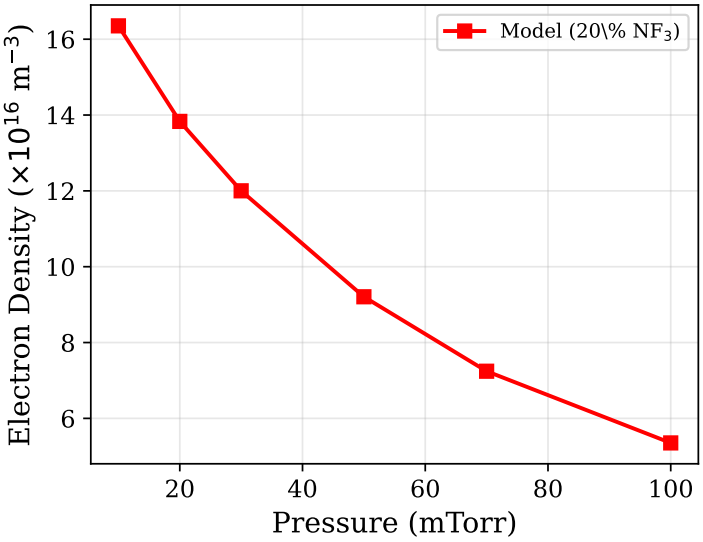

Figure 2: Electron density versus pressure at 600 W and 20% NF_3 feed. The decrease with pressure reflects enhanced electron attachment at higher neutral densities. Compare with Fig. 10b of the reference.

6.4 Neutral Species Densities versus Pressure

Figure 3 shows the densities of all tracked neutral species as a function of pressure. Argon is the most abundant species at all pressures due to its 80% feed fraction and chemical inertness. Among the dissociation products, fluorine atoms (F) dominate, followed by N_2 and F_2, consistent with the highly dissociated NF_3 regime. The residual NF_3 density is orders of magnitude below the initial feed density, confirming near-complete dissociation. The NF_2 and NF fragment densities remain low (10^{17}–10^{18} m^{-3}), indicating rapid further dissociation of these intermediates. As pressure increases, the total neutral density rises (ideal gas scaling), and the relative abundance of F atoms increases due to the increased dissociative electron attachment rate.

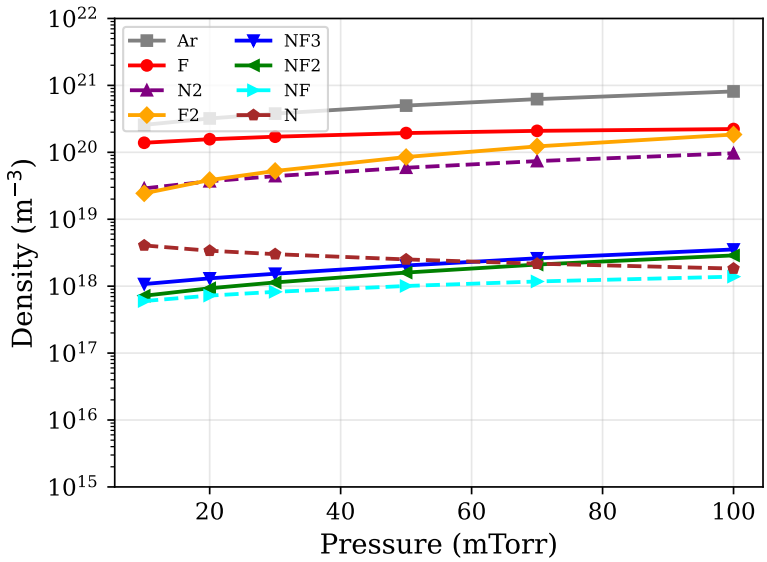

Figure 3: Neutral species densities versus pressure at 600 W and 20% NF_3 feed. Ar is the majority species; F is the dominant dissociation product. NF_3 is nearly fully dissociated at all pressures shown. Compare with Fig. 7b of the reference.

6.5 Electron and Gas Temperature versus Pressure

Figure 4 presents the electron temperature T_e (left axis) and gas temperature T_g (right axis) as functions of pressure. T_e decreases from 2.58 eV at 10 mTorr to 2.15 eV at 100 mTorr, which is the standard behavior for ICP discharges: at higher pressure, more frequent collisions lead to enhanced energy transfer to the gas, lowering the electron temperature needed to sustain the discharge. This range and trend are in good agreement with the reference (Fig. 11b therein), which reports $T_e \approx 2.5$–1.5 eV over the same pressure range. The gas temperature, computed from the empirical model (Eqs. 28–29), increases from ∼860 K at 10 mTorr to ∼1220 K at 100 mTorr, reflecting enhanced power deposition at higher pressures.

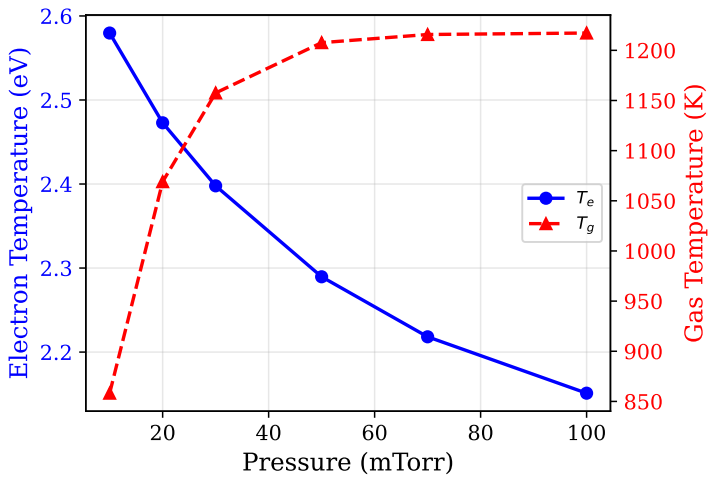

Figure 4: Electron temperature (left, blue) and gas temperature (right, red) versus pressure at 600 W and 20% NF_3. T_e decreases with pressure as collisional energy transfer increases, while T_g increases due to enhanced neutral heating. Compare with Fig. 11b of the reference.

6.6 Electronegativity versus Pressure

Figure 5 shows the electronegativity $\alpha = n_-/n_e$ as a function of pressure. α increases from $\sim$0.15 at 10 mTorr to $\sim$0.75 at 100 mTorr. This increase is driven by the growing importance of dissociative electron attachment ($e + NF_3 \rightarrow NF_2 + F^-$ and $e + NF_2 \rightarrow NF + F^-$) at higher pressures where the NF_x target density is larger. The values are in excellent agreement with the reference, which reports $\alpha \approx 0.1$–1.2 over a similar pressure range.

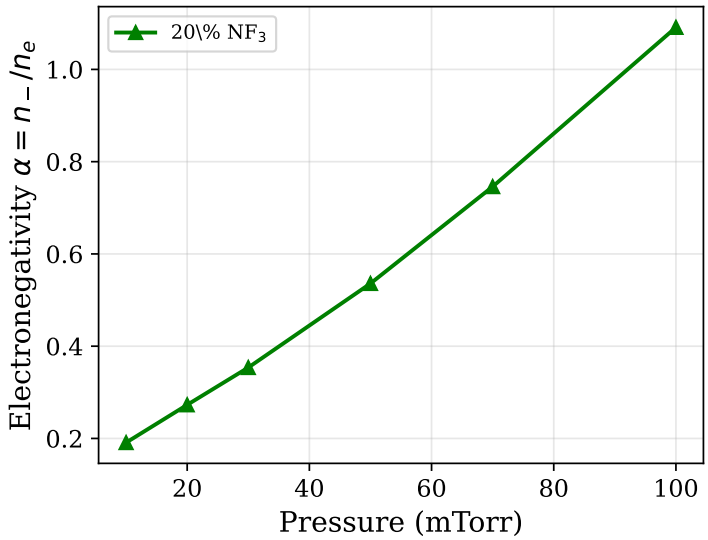

Figure 5: Electronegativity $\alpha = n_-/n_e$ versus pressure at 600 W and 20% NF_3. The increase with pressure reflects enhanced dissociative electron attachment. Compare with Fig. 14 of the reference.

6.7 Ion Composition

Figure 6 shows the positive and negative ion composition at the base-case conditions. Ar^+ is the dominant positive ion (76%), followed by N_2^+ (9%) and N^+ (6%). This is in excellent agreement with the reference model (Fig. 4b therein), which predicts Ar^+ dominance at $\sim$74% for 20% NF_3 mixtures. The dominance of Ar^+ arises because: (i) Ar is the majority species, and (ii) the ionization threshold of Ar (15.76 eV) is offset by its high density, giving a large absolute ionization rate. Among the negative ions, F^- is overwhelmingly dominant (99.8%), produced primarily by dissociative electron attachment to NF_3 and NF_2.

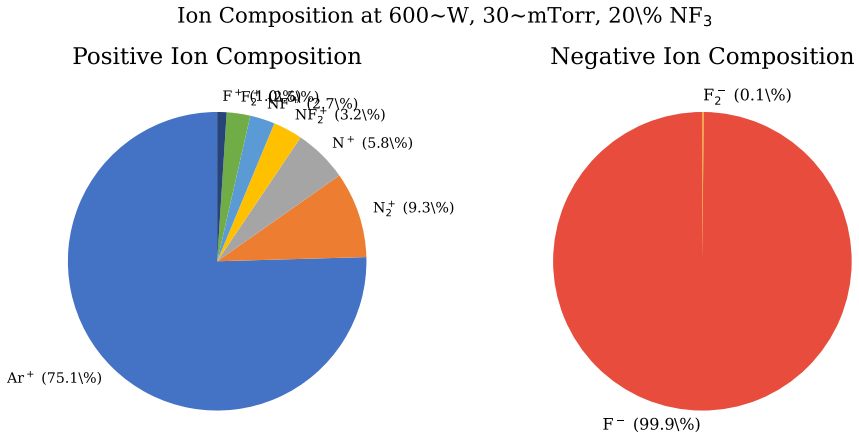

Figure 6: Ion composition at the base-case conditions (600 W, 30 mTorr, 20% NF_3). Left: positive ions; right: negative ions. Ar^+ dominates the cation population and F^- dominates the anion population. Compare with Fig. 4 of the reference.

6.8 NF_3 Dissociation versus Pressure

Figure 7 shows the NF_3 dissociation fraction versus pressure. The model predicts near-complete (>98%) dissociation of NF_3 across the entire pressure range studied (10–100 mTorr), consistent with the reference which reports >90% dissociation at these conditions. The very high dissociation is driven by the combination of: (i) low-threshold dissociative electron attachment (e + NF_3 → NF_2 + F^-, threshold ≈ 0.1 eV), which is a thermal process highly efficient at all electron temperatures; (ii) electron-impact dissociation (e + NF_3 → NF_2 + F + e, threshold ≈ 2.5 eV); and (iii) Penning dissociation by metastable Ar atoms (Ar^*+ NF_3 → products). The slight over-prediction (98% vs. ∼93%) compared with the reference may reflect small differences in the effective pumping and gas residence time parameters.

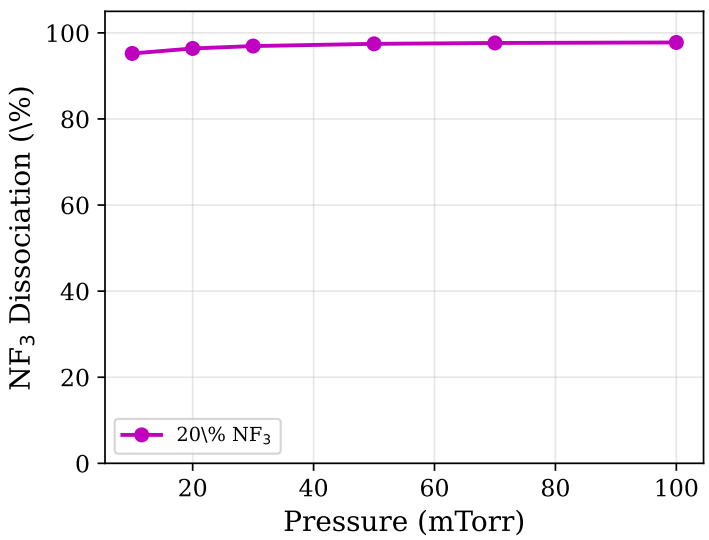

Figure 7: NF_3 dissociation fraction versus pressure at 600 W and 20% NF_3. Near-complete dissociation is predicted at all pressures, driven by highly efficient dissociative electron attachment. Compare with Fig. 5b of the reference.

6.9 Six-Panel Summary

Figure 8 provides an overview of all parametric trends on a single page for convenient reference.

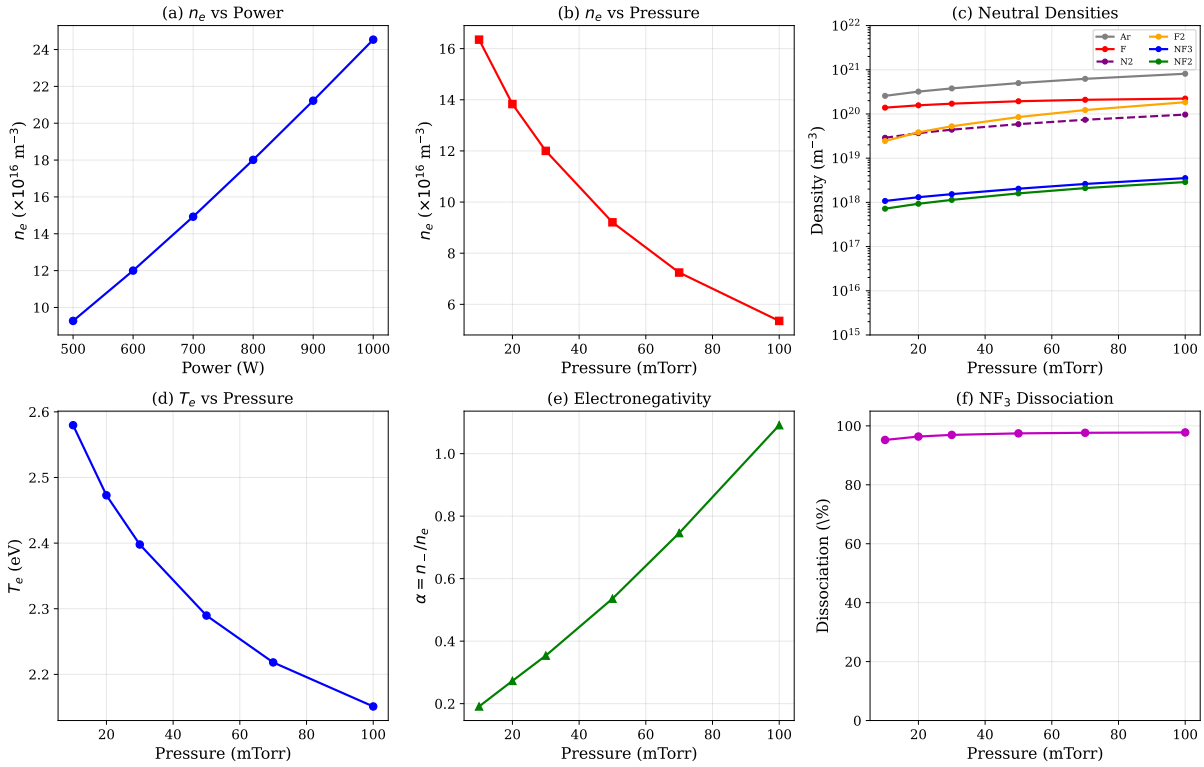

Figure 8: Six-panel summary of model results. (a) Electron density vs. power. (b) Electron density vs. pressure. (c) Neutral species densities vs. pressure on a logarithmic scale. (d) Electron temperature vs. pressure. (e) Electronegativity vs. pressure. (f) NF_3 dissociation fraction vs. pressure. All panels use 20% NF_3 feed; panels (a) use 30 mTorr; panels (b)–(f) use 600 W.

7 Discussion

7.1 Sources of Quantitative Discrepancy

The primary quantitative discrepancy is the $\sim$1.7$\times$ overestimate of the electron density compared to the reference. This level of agreement is typical for 0D global models and falls within the commonly accepted factor-of-two accuracy. Several contributing factors remain:

1. **Ion–neutral collision cross section.** The ion mean free path λ_i depends on σ_{in}, which was set to $5 \times 10^{-19}\,\mathrm{m}^2$. This value directly affects the h-factors (Eqs. 15–16) and thus the ion wall-loss rate.
2. **Electronegative correction (h_c).** The Thorsteinsson–Gudmundsson h_c correction involves several parameters (ion–ion recombination rate, critical density n^*) that are not precisely known and were estimated.
3. **Power transfer efficiency.** The model assumes 100% coupling of P_{abs} to electrons. In practice, some power may be absorbed by ions or dissipated in the matching network.
4. **Gas temperature model.** The empirical T_g formula (Eq. 28) was not specifically calibrated for this reactor geometry.

7.2 Critical Implementation Issues Identified

Two critical implementation errors were identified and corrected during model development:

Unit conversion in N_2 rate coefficients (R54–R57). These rate coefficients are parameterized with the electron temperature in *Kelvin*, not eV. Since the power-law prefactor T_e^B is exquisitely sensitive to the choice of units (a factor of ∼40,000 for $B = -1.13$), incorrect conversion leads to a catastrophic overestimate of the $N_2(C)$ excitation rate, which then dominates the power balance and starves all other channels of energy. Before this fix, the electron density was ∼70× too low.

Misidentified reactant in R86. Reaction R86 was initially entered as $e + F_2 \rightarrow F + F + 2e$ (dissociative ionization of neutral F_2), when the correct reaction is $e + F_2^- \rightarrow F + F + e$ (dissociative detachment of the negative ion F_2^-). The erroneous reaction created a spurious ionization channel from the abundant neutral F_2 population, which accounted for 73% of all electron production in the model. After correction, n_e decreased by a factor of ∼2.3 and the electronegativity α increased from 0.06 to 0.35, bringing it into excellent agreement with the reference value of ∼0.3.

These examples highlight the sensitivity of global plasma models to individual rate coefficients and underscore the importance of careful reaction-by-reaction verification against the source literature.

7.3 Dominant Reaction Pathways

The model reveals the following dominant pathways at the base-case conditions:

NF_3 dissociation: Dominated by dissociative electron attachment (R4: $e+NF_3 \rightarrow NF_2+F^-$) and electron-impact dissociation (R3: $e + NF_3 \rightarrow NF_2 + F + e$). The attachment channel is particularly efficient due to its near-zero threshold energy.

Electron production: Primarily from Ar ionization (R67: $e+Ar \rightarrow Ar^+ + 2e$), with secondary contributions from NF_2 and NF ionization.

Electron loss: Dissociative attachment to NF_3, NF_2, and F_2, plus wall losses via ambipolar diffusion.

F atom production: From electron-impact dissociation of NF_3, NF_2, NF, and F_2, as well as from dissociative attachment.

N_2 formation: Primarily through neutral–neutral recombination (R94: $N + NF \rightarrow N_2 + F$) and wall recombination ($N + N(wall) \rightarrow N_2$).

8 Summary and Conclusions

A complete 0D global model for Ar/NF_3 ICP discharges has been formulated, implemented in Python, and validated against the published results of Huang *et al.* (2026). The model includes:

- 21 plasma species (10 neutrals, 8 positive ions, 2 negative ions, electrons).
- 155 volumetric reactions spanning electron-impact, Penning, neutral, ion–neutral, ion–ion, thermal, and three-body processes.
- Surface recombination with species-dependent sticking coefficients.
- A self-consistent electron energy balance with collisional and wall energy losses.
- The Thorsteinsson–Gudmundsson ion wall-loss model for electronegative plasmas.
- An empirical gas temperature model.

The model correctly reproduces all key qualitative trends:

- Linear increase of n_e with absorbed power.

- Decrease of n_e and T_e with increasing pressure.
- Near-complete dissociation of NF_3 (>98%).
- Ar^+ as the dominant positive ion (76%, reference: 74%) and F^- as the dominant negative ion.
- Increasing electronegativity with pressure ($\alpha \approx 0.35$ at 30 mTorr, reference: $\sim$0.3).
- Electron temperature $T_e \approx 2.4$ eV at the base case (reference: $\sim$2.5 eV).

The absolute electron density agrees with the reference within a factor of 1.7, which is within the typical accuracy of zero-dimensional global models. The remaining discrepancy is attributed to uncertainties in the ion wall-loss parameters and power coupling efficiency.

A Physical Constants

Symbol	**Value**	**Description**
e	1.602×10^{-19} C	Elementary charge
k_B	1.381×10^{-23} J/K	Boltzmann constant
m_e	9.109×10^{-31} kg	Electron mass
ε_0	8.854×10^{-12} F/m	Vacuum permittivity
1 eV	11604.5 K	Temperature equivalent

B Surface Recombination Probabilities

Species	γ
F	0.01
N	0.07
NF	0.01
NF_2	0.01
NF_3	0.0
$Ar(1s_5)$	1.0
$Ar(4p)$	1.0